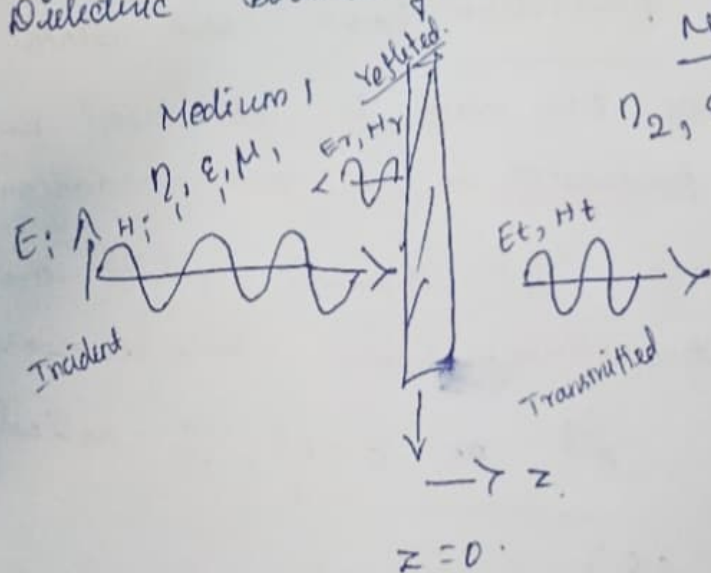


Group velocity.

- i) Group velocity is defined as a rate at which the wave propagates through the waveguide.
- ii) Group velocity is the rate at which modulated envelope travels as compared to the carrier alone. This modulated wave travels through the wave guide.
- iii) The equation for group velocity is

$$v_g = \frac{d\omega}{dB}$$

Normal Incidence of a wave at a Dielectric Boundary.



No valence. $\frac{e^-}{s}$
 $\downarrow$
 $\sigma \approx 0$
 $\gamma \approx 0$
 elec

→ Reflection coefficient ' Γ '

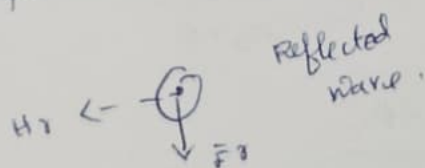
→ Transmission "

→ SWR.

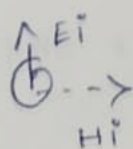
Normal incidence at a plane dielectric.

Boundary.

→ when an electromagnetic wave is incident normally on the surface of a dielectric medium, then the part of the incident power is transmitted & part is reflected.



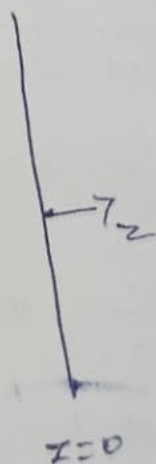
Reflected wave.



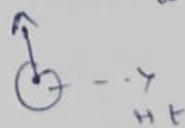
Incident wave.

M-1

$\epsilon_1, \mu_1, \eta_1$



Transmitted wave.



Dielectric

M-2

$\epsilon_2, \mu_2, \eta_2$

Incident wave:-

$$E_i(z) = E_{i0} e^{-j\beta_1 z} \hat{a}_x$$

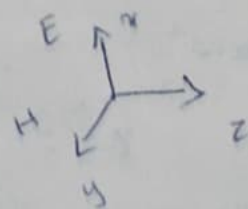
$$H_i(z) = \frac{E_{i0}}{\eta} e^{-j\beta_1 z} \hat{a}_y$$

For the reflected wave.

$$E_r(z) = E_{r0} e^{j\beta_1 z} \hat{a}_x$$

opposite direction.

$$H_r(z) = \frac{E_{r0}}{\eta} e^{j\beta_1 z} \hat{a}_y$$



$$\frac{H_{i0}}{\eta} e^{-j\beta_1 z}$$

$$\eta = \frac{E}{H}$$

Transmitted wave, (happens in m_2)

$$E_t(z) = E_{t0} e^{-j\beta_2 z} \hat{a}_x$$

$$H_t(z) = \frac{E_{t0}}{\eta_2} e^{-j\beta_2 z} \hat{a}_y$$

* Reflection coefficient (Γ)

$$\Gamma \Rightarrow \frac{\text{Reflected wave}}{\text{Incident wave}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\Gamma = \frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$|\Gamma| = \frac{S-1}{S+1} ; \text{ in terms of SWR.}$$

$$-1 \leq \Gamma \leq 1$$

Transmission coefficient: - (τ)

Ratio of transmitted wave to incident wave.

$$\tau = \frac{E_t}{E_i} = \frac{2\eta_2}{\eta_1 + \eta_2}$$

$$1 + \Gamma = \tau$$

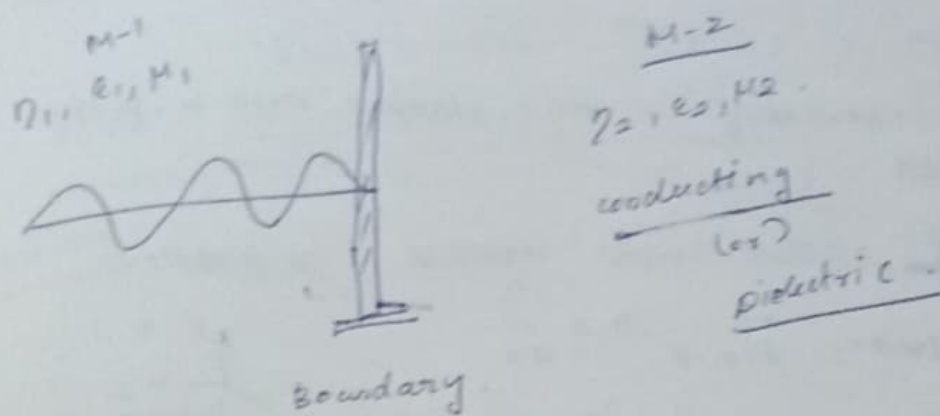
(Def)

standing wave Ratio: (S)

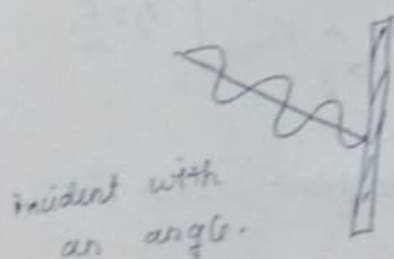
$$S = \frac{E_{max}}{E_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$1 < S < \infty$$

Normal incidence of a wave.

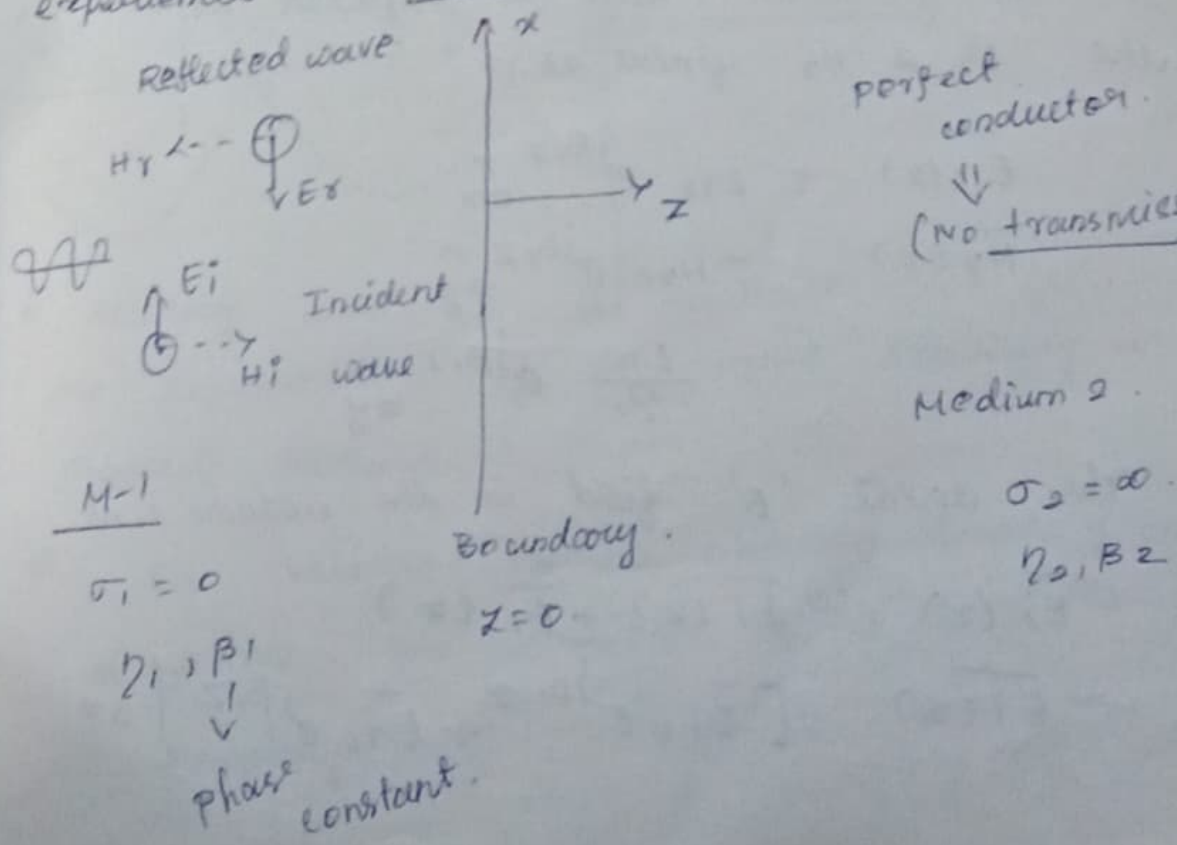


oblique Incidence.



Normal Incidence at a plane conducting boundary.

when an electromagnetic wave is incident normally on a plane conducting boundary, it experiences a reflection.



⇒ Assumption:

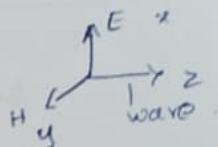
→ Incident wave (E_i, H_i) travels in a lossless medium 1 ($\sigma=0$)

→ Boundary is an interface with a perfect conductor ($\sigma=\infty$)

→ Incident wave travels in z -direction.

$$\vec{E}_i(z) = \vec{E}_{i0} e^{-j\beta_1 z} \hat{a}_x$$

$$\vec{H}_i(z) = \vec{H}_{i0} e^{-j\beta_1 z} \hat{a}_y$$



$$\frac{E_{i0}}{\eta_1} e^{-j\beta_1 z} \hat{a}_y \quad \left| \quad \eta = \frac{E}{H} \right.$$

The Poynting vector of the incident wave is,

$$\vec{P}_i(z) = \vec{E}_i(z) \times \vec{H}_i(z)$$

→ Since there is no wave transmitted across the boundary.

$$\text{Both } E_2 \text{ \& } H_2 = 0$$

→ The incident wave is reflected with the E_r & H_r fields as.

$$\vec{E}_r(z) = E_{r0} e^{j\beta_1 z} \hat{a}_x$$

$$\vec{H}_r(z) = H_{r0} e^{j\beta_1 z} \hat{a}_y$$

$$= \frac{E_{r0}}{\eta_1} e^{j\beta_1 z} \hat{a}_y$$

The total 'E' field in the medium 1 is

$$\vec{E}_t(z) = \vec{E}_i(z) + \vec{E}_r(z)$$

$$\vec{E}_t(z) = \left[\vec{E}_{i0} e^{-j\beta_1 z} + \vec{E}_{r0} e^{j\beta_1 z} \right] \hat{a}_x$$

At $z=0$.

$$E_i(z) = (\bar{E}_{i0} + \bar{E}_{i0}) \hat{a}_x = 0$$

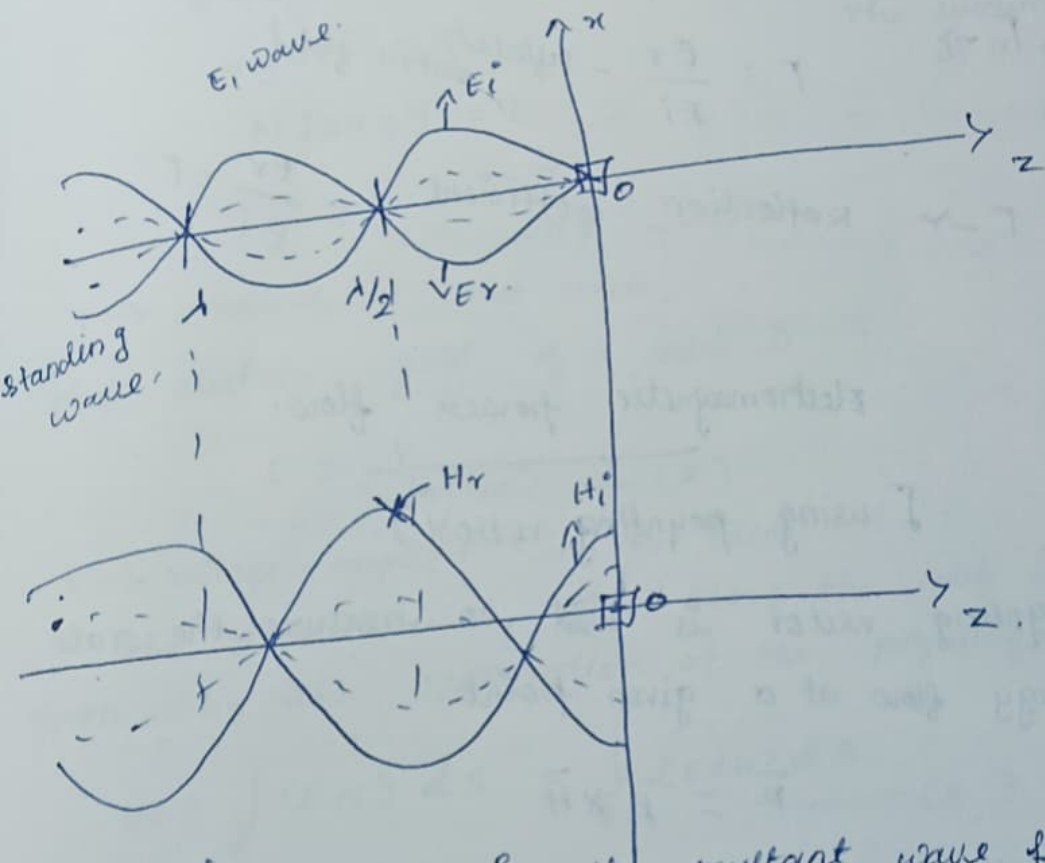
$$\bar{E}_{i0} = -\bar{E}_{r0} \text{ at } z=0$$

$$E_i(z) = j \omega E_{i0} \sin \beta_1 z \cdot \hat{a}_x \quad (\text{im})$$

Similarly,

$$H_i(z) = \frac{2E_{i0}}{\eta_1} \cos \beta_1 z \hat{a}_y \quad (\text{Real})$$

standing wave:



* Standing wave is the resultant wave from the superposition of two waves travelling in opposite direction.

* E_i vanishes on the conducting boundary and at points are multiples of $1/2$ from the boundary.

* H_1 is maximum on the conducting boundary.

def: standing wave ratio (SWR):-

$$S = \frac{E_{max}}{E_{min}} ; 1 < |S| < \infty$$

$$S = \frac{1+|\Gamma|}{1-|\Gamma|} ; -1 \leq |\Gamma| \leq 1$$

$\Gamma \rightarrow \Gamma = \frac{E_r}{E_i}$ - Reflected electric field.

$\Gamma \rightarrow$ Reflection coefficient $= \frac{E_r}{E_i} = \Gamma$

Electromagnetic power flow:

[using Poynting vector].

Poynting vector is used to measure the rate of energy flow at a given point.

$$\vec{P} = \vec{E} \times \vec{H}$$

Applications:

1. TO measure the power flow.
2. TO find the direction of power.

of $\vec{H}$ is continuous across the boundary.

unit - 5
Electromagnetic waves.

$\Rightarrow$ plane wave in a lossless medium.

A medium is lossless if the conductivity is zero (i.e.) $\sigma = 0$

Ex: Air & Vacuum space have no conductivity.

A lossless medium does not dissipate EM energy due to poor conduction.

(i.e.) No power loss.

In general,

$$\alpha = \omega \sqrt{\frac{\mu \epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} - 1 \right)}$$

Attenuation constant.

As $\sigma = 0$ (conductivity)

$$\alpha = \omega \sqrt{\frac{\mu \epsilon}{2} (\sqrt{1+0} - 1)}$$

$$\alpha = \omega \sqrt{\frac{\mu_0}{2}} (0) = 0$$

β (phase shift constant):-

$$\beta = \omega \sqrt{\frac{\mu \epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} + 1 \right)}$$

since $\sigma = 0,$

$$\beta = \omega \sqrt{\frac{\mu \epsilon}{2} (\sqrt{1+1})} \Rightarrow \omega \sqrt{\frac{\mu \epsilon}{2} \times 2}$$

$$\beta = \omega \sqrt{\mu \epsilon}$$

v : - wave velocity)

$$v = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{\mu \epsilon}}$$

$$v = \frac{1}{\sqrt{\mu \epsilon}} \quad \text{m/s}$$

λ : wave length:

$$\lambda = \frac{2\pi}{\beta} \quad ; \quad \text{m}$$

η : Intrinsic Impedance:

$$\eta = \sqrt{\frac{\mu/\epsilon}{1 + \sigma/j\omega\epsilon}} \quad ; \quad \sigma = 0$$

$$\eta = \sqrt{\mu/\epsilon} \quad ; \quad \Omega$$

phase value (θ):

$$\tan \theta = \frac{\sigma}{\omega\epsilon} = 0$$

$$\theta = 0$$

Both $\vec{E}$ & $\vec{H}$ are in same phase.

plane waves in free space.

$$\sigma = 0; \quad \epsilon_r = 1; \quad \mu_r = 1$$

* characteristics of plane waves in free space.

$$i) \quad \alpha = 0$$

$$ii) \quad \beta = \omega \sqrt{\mu \epsilon}$$

$$= \omega \sqrt{\mu_0 \epsilon_0 \epsilon_r \mu_r}$$

$$\therefore \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$\beta = \omega \sqrt{\mu_0 \epsilon_0}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\text{iii) } v = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{\mu_0 \epsilon_0}}$$

$$= \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s} = c$$

EM waves travel in free space at the speed of light.

* Intrinsic impedance.

$$\eta = \eta_0 = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.85 \times 10^{-12}}}$$

$$\eta_0 = 120\pi \approx 377 \Omega$$

plane waves in lossy
mediums.

plane waves in medium.

→ carries information from one place to another over a medium.

Ex: Radio waves, TV signals etc.

characteristics:

propagation constant (γ)

Attenuation constant (α)

phase shift (β)

velocity of EM waves (v)

Wave length (λ)

Types of medium:

i) lossless medium ($\sigma=0$)

ii) lossy medium.

low loss
dielectric

$$(\sigma \neq 0)$$

Good conductor.

$$(\sigma = \infty)$$

(plane waves in lossy medium.)

Wave propagation in lossy dielectrics.

→ It is a partially conducting medium → Imperfect conductor.

i.e.) $\sigma \neq 0$ (or) finite value.

→ As wave propagates, there is a power.

Wave equation in lossy dielectric.

consider a lossy dielectric medium with charge free condition.

$$\boxed{\rho = 0}$$

Maxwell's Equations:

$$\nabla \times \vec{H} = (\sigma + j\omega\epsilon) \vec{E} \quad \text{--- (1)}$$

$$\nabla \times \vec{E} = -j\omega\mu \vec{H} \quad \text{--- (2)}$$

$$\nabla \cdot \vec{D} = 0 \quad \Rightarrow \quad \nabla \cdot \vec{E} = 0 \quad \text{--- (3)}$$

$$\nabla \cdot \vec{B} = 0 \quad \Rightarrow \quad \nabla \cdot \vec{H} = 0 \quad \text{--- (4)}$$

consider Maxwell's (2) equation.

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}$$

Taking curl on both sides.

$$\nabla \times (\nabla \times \vec{E}) = -j\omega\mu (\nabla \times \vec{H}) \quad \text{--- (5)}$$

From the identity,

$$\nabla \times (\nabla \times \vec{E}) = \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

From equation (3), $\nabla \cdot \vec{E} = 0$

$$\nabla \times \nabla \times \vec{E} = -\nabla^2 \vec{E} \quad \text{--- (6)}$$

sub eqn (1) & (6) in (5),

$$-\nabla^2 \vec{E} = -j\omega\mu (\sigma + j\omega\epsilon) \vec{E}$$

$$\nabla^2 \vec{E} = \gamma^2 \vec{E} \quad \text{--- (7)}$$

where,

$$\gamma^2 = j\omega\mu (\sigma + j\omega\epsilon) \quad \text{--- (8)}$$

The eq (7) becomes.

$$\boxed{\nabla^2 \vec{E} - \gamma^2 \vec{E} = 0}$$

This is called homogeneous vector wave equation.

* To find γ , α , β & θ :-

$$\gamma^2 = j\omega\mu (\sigma + j\omega\epsilon); \quad \gamma = \alpha + j\beta$$

$$= j\omega\mu\sigma + j^2\omega^2\mu\epsilon; \quad \gamma^2 = (\alpha + j\beta)^2$$

$$\gamma^2 = j\omega\mu\sigma - \omega^2\mu\epsilon; \quad = \alpha^2 + j^2\beta^2 + 2j\alpha\beta$$

--- (A)

$$\gamma^2 = (\alpha^2 - \beta^2) + j2\alpha\beta \quad \text{--- (B)}$$

compare real & imaginary part of A & B

$$\left. \begin{aligned} \alpha^2 - \beta^2 &= \frac{-\omega^2 \mu \epsilon}{2} \\ 2\alpha\beta &= \omega \mu \sigma \end{aligned} \right\} \begin{aligned} a^2 &= x + jy \\ |a^2| &= \sqrt{x^2 + y^2} \end{aligned}$$

$$|\gamma^2| = \sqrt{(\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2}$$

$$\begin{aligned} |\gamma^2| &= \sqrt{(\alpha^2 - \beta^2)^2 + (2\alpha\beta)^2} \\ &\Rightarrow \sqrt{\alpha^4 + \beta^4 - 2\alpha^2\beta^2 + 4\alpha^2\beta^2} \\ &\Rightarrow \sqrt{\alpha^4 + \beta^4 + 2\alpha^2\beta^2} \\ |\gamma^2| &= \sqrt{(\alpha^2 + \beta^2)^2} = \alpha^2 + \beta^2 \end{aligned} \quad \left\{ \begin{aligned} &\Rightarrow \sqrt{(-\omega^2 \mu \epsilon)^2 + (\omega \mu \sigma)^2} \\ &\Rightarrow \sqrt{\omega^4 \mu^2 \epsilon^2 + \omega^2 \mu^2 \sigma^2} \\ &\Rightarrow \sqrt{\omega^2 \mu^2 (\omega^2 \epsilon^2 + \sigma^2)} \\ |\gamma^2| &= \mu \omega \sqrt{\omega^2 \epsilon^2 + \sigma^2} \\ |\gamma^2| &= \omega \mu \sqrt{\sigma^2 + \omega^2 \epsilon^2} \end{aligned} \right.$$

$$\alpha^2 + \beta^2 = \omega \mu \sqrt{\sigma^2 + \omega^2 \epsilon^2}$$

$$\alpha^2 - \beta^2 = -\omega^2 \mu \epsilon$$

$$2\alpha^2 = \omega \mu \sqrt{\sigma^2 + \omega^2 \epsilon^2} - \omega^2 \mu \epsilon$$

$$\alpha^2 = \frac{\omega^2 \mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right]$$

$$\alpha^2 \Rightarrow \frac{\omega^2 \mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right]$$

Attenuation
constant

$$\left\{ \begin{aligned} \alpha &= \omega \sqrt{\frac{\mu \epsilon}{2}} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right)^{1/2} \\ \alpha &= \omega \sqrt{\frac{\mu \epsilon}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right]^{1/2} \end{aligned} \right.$$

Attenuation
constant (B)

similarly;

$$\beta = \omega \sqrt{\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} + 1 \right]}$$

phase velocity;

$$v_p = \frac{\omega}{\beta}$$

$$= \frac{\omega}{\omega \sqrt{\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} + 1 \right]}}$$

$$v_p = \left[\sqrt{\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} + 1 \right]} \right]^{-1}$$

Wave propagation in good conductor.

$$\sigma = \infty$$

$$\frac{\sigma}{\omega \epsilon} \gg 1$$

As wave propagates in conductor, there is very high power loss (∞ conduction).

$$\alpha = \omega \sqrt{\frac{\mu \epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right)}$$

since $\frac{\sigma}{\omega \epsilon} \gg 1$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon}{2} \left(\sqrt{\left(\frac{\sigma}{\omega \epsilon}\right)^2} - 1 \right)}$$

$$= \omega \sqrt{\frac{\mu \epsilon \sigma}{2 \omega \epsilon}} - 1$$

$$= \sqrt{\frac{\omega \mu \sigma}{2}}$$

$$\alpha = \sqrt{\frac{2\pi f \mu \sigma}{2}}$$

$$\alpha = \sqrt{\pi f \mu \sigma}$$

$$\alpha = \sqrt{\frac{\omega \mu \sigma}{2}}$$

$$= \sqrt{\frac{2\pi f \mu \sigma}{2}}$$

$$\frac{\sin \theta}{\sin \theta} \quad \beta = \sqrt{\pi f \mu \sigma}$$

$\alpha = \beta$ in good conductor.

velocity:

$$v_p = \frac{\omega}{\beta}$$

$$= \frac{2\pi f}{\sqrt{\pi f \mu \sigma}} = \frac{\sqrt{2\pi f} \cdot \sqrt{2\pi f}}{\sqrt{\pi f \mu \sigma}}$$

$$v_p = \gamma \frac{2\omega}{\mu \sigma} \Rightarrow 2\sqrt{\frac{\pi f}{\mu \sigma}}; \text{ (m/s)}$$

Depth of penetration (or) skin depth.

A high frequency EM wave is attenuated very rapidly as it propagates in a good conductor.

$\rightarrow$ The distance 's' through which the amplitude of a travelling plane wave decreases by a factor of e^{-1} or 0.368 is called skin depth (δ).

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$